Karan Srinivas

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EDUCATION

Northeastern University **Boston, MA**
Master of Science in Computer Engineering (GPA-3.8) Jan 2024-May 2026
Coursework – Object Oriented Design, Data Structures & Algorithms, Generative AI, Operating System, High Performance Computing, AI Agent Infrastructure, Data Science Engineering, Machine Learning

WORK EXPERIENCE

Software Engineer Intern/Co-op, Vicor Corporation **Andover, MA | Jun 2025 – Dec 2025**

- **Developed AI automated API workflows (Python, Flask, Azure OpenAI, Azure Functions)** autonomously orchestrating power-module selection and converter simulation reducing validation turnaround by **50%**
- Developed automated **PyTest** and **CppUTest** suites **for C++/Python CI pipelines** and **HIL** benches, using **GitHub Copilot** and **Claude AI workflows** and improve **I²C/PMBus** validation and **regression** detection by **40%**
- Implemented cloud automation for chip and power-module simulations using **Docker**, **Azure Container Apps**, **Azure (Functions, Monitor)**, reducing **deployment time** and improving analysis efficiency by **35%**
- **Developed an AI optimized data layer (Python, LangChain, Azure Cosmos DB, Azure OpenAI with GPT 4.0)** with versioned schemas and **3K+ automated queries** autonomously storing, retrieving, and analyzing Vicor power-chip simulation results accelerating reporting by **25%**

Graduate Research Assistant, Northeastern University **Boston, MA | May 2024 – May 2025**

- Deployed **student feedback analysis system (Python, LangChain, AWS Bedrock with GPT 4.0)** under **Prof. Gunar Schirner** to autonomously analyze **5,000+ student submissions**, cutting manual review by **40%**
- Built an **AI-automated pipeline (S3 → Lambda → Bedrock GPT-4.0 → RDS)** autonomously orchestrating feedback ingestion and analysis for **5,000+ students**, boosting throughput by **25%**
- Developed an **agentic RAG app (Django, React, LangChain, AWS Bedrock with GPT 4.0)** where LLM agents autonomously route and summarize feedback across **5,000+ students 3x faster delivery**

Software Engineer, Tata Consultancy Services **Bangalore, IN | Aug 2021 – Dec 2023**

- Developed **Wire, NACHA,** and **ACH** payment features in **Java/J2EE**, **Spring Boot**, and **PL/SQL** scaled to 100K+ daily transactions, improved throughput by 30%, and maintained compliance **SWIFT/NACHA/PCI DSS**
- Designed and provisioned **SOA-based distributed microservices** on **AWS (ELB, EC2, S3, EKS, Kubernetes)** using **Terraform** for full **IaC automation**, handling **1M queries daily API queries** and achieving **75% performance**
- Developed a **real-time NLP chatbot** backend using **Spring Boot, TCP socket based I/O**, **Java** and **Apache OpenNLP** integrating **Kafka** and **AWS SQS** for event-driven messaging and increasing **user engagement by 50%**
- Built a mobile banking iOS application using **Swift** and **Xcode**, providing seamless customer experiences for account management, transaction tracking, and fund transfers
- Developed and automated **XCTest**, **Selenium, Mockito** and **JUnit test** suites using **applied TDD** across all banking modules and integrated into **Jenkins CI/CD pipelines** and reduced production bugs by 40%

ACADEMIC PROJECTS

Credit Card Cost Management (React JS, Django, CI/CD, Docker, Python) **Jan 2025 – Apr 2025**

- **Built an AI-driven spending analysis engine** (**Python, Open AI(GPT 3.5), MongoDB**) autonomously categorizing transactions and surfacing personalized spending insights across **multi-card portfolio**
- **Integrated JWT-secured REST APIs** (**Django, Python, MongoDB**) with an **OpenAI agent layer** autonomously routing user queries and triggering bill payment workflows and improving transaction efficiency by **35%**

TECHNICAL SKILLS

Programming Languages	Java, C++, C#, JavaScript, TypeScript, Python, SQL, C, React.js, Node.js,
Databases and Frameworks	MySQL, MongoDB, Django, TensorFlow, IBM DB2, Spring Boot, Hibernate
Cloud/IaaC/DevOps	AWS (EC2, RDS, S3, CloudWatch, Lambda), Airflow, GCP, Azure, Snowflake
Software and CI/CD Tools	GitHub Actions, Packer, Docker, DevSecOps, SaaS, Maven, Jenkins, IaC

PUBLICATIONS

Demand forecasting and Route Optimization in Supply chain industry | Publication Link **October 2021**
Published IEEE research on ML-driven demand forecasting and route optimization (ARIMA, Simulated Annealing), applying time series analysis and data analytics to optimize supply chain operations and improving forecast accuracy